

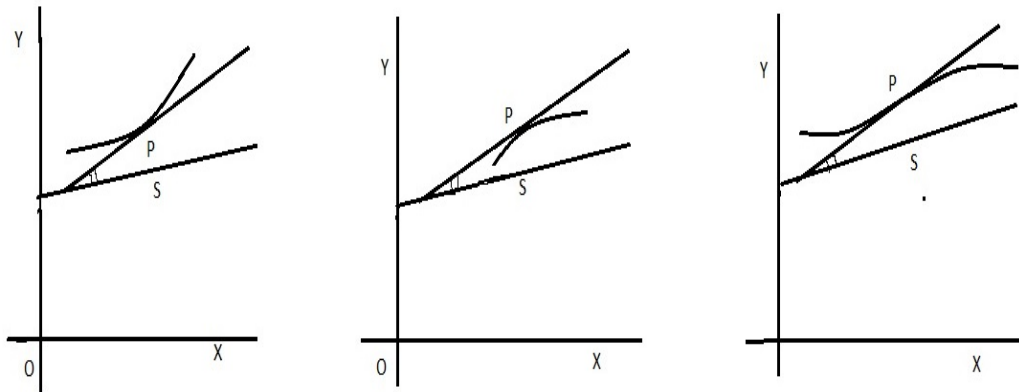
Lecture: 17 Curve Tracing

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Concavity and Convexity

Let P be a given point on a curve and S a given straight line which does not pass through P . Then, the curve is said to be concave or convex at P with respect to S , according as a sufficiently small arc containing P lies entirely within or without the acute angle formed by S and the tangent to the curve at P .

If the curve on one side of P is concave, and on the other side of P



convex to any line S , the curve evidently crosses its tangent at P . Such a point is called a *point of inflexion*.

A test of concavity or convexity or point of inflexion

1. A curve is convex or concave at P to the axis of x according as $y \frac{d^2y}{dx^2}$ is positive or negative at P .
2. There is a point of inflexion at P , if

$$\frac{d^2y}{dx^2} = 0, \text{ but } \frac{d^3y}{dx^3} \neq 0.$$

Multiple Points

A point of inflexion is a singular point (i.e. an unusual point) on the curve at which the tangent crosses the curve is the point of inflexion.

A point through which more than one branches of a curve pass is called a multiple point on the curve.

A point on a curve which two branches of the curve passes through it; a triple point if three branches pass through it and so on. If r branches pass a point, it is called a multiple point of the r th order.

Node: If the two branches through a double point on a curve are real, and the tangents to them are not coincident, the double point is called a node.

Cusp: If the two tangents are coincident, the point is called a cusp.

Conjugate Point: If there are no real points on the curve in the neighborhood of a point P on the curve. P is called a conjugate point (or an isolated point).

Let $f(x, y) = 0$ be an equation to a curve then a double point will be a node, cusp, or conjugate point according as there

$$\left(\frac{\partial^2 f}{\partial x \partial y} \right)^2 >, =, < \left(\frac{\partial^2 f}{\partial x^2} \right) \left(\frac{\partial^2 f}{\partial y^2} \right).$$

Cartesian Curve Tracing

1. Symmetrical

- **Symmetric about x -axis:** If all the powers of y are even.
Example. $y^2 = 4ax$.
- **Symmetric about y -axis:** If all the powers of x are even.
Example. $x^2 = 4ay$.

- **Symmetric about the line $y = x$:** Replace $x \mapsto y$, if the equation is unchanged then symmetric about line $y = x$.

Example. $x^3 + y^3 = 3axy$.

Example.

$$x^2 + y^2 + x^4y = 0.$$

$$x \mapsto y \quad y^2 + x^2 + y^4x = 0.$$

$\Rightarrow$ Not symmetrical about line $y = x$.

- **Symmetric about line $y = -x$:** Replace $x \mapsto -y$ and $y \mapsto -x$, if the equation remains unchanged then curve is symmetrical about line $y = -x$.
- **Symmetric about opposite quadrant:** Replace $x \mapsto -x$ and $y \mapsto -y$, if equation remains unchanged then symmetrical about opposite quadrant.

Example. $y = x^2$.

2. **Passing through $(0,0)$:** The curve passes through the origin, if the equation does not contain constant terms.
Example. $y^2 = 4ax$ passes through the origin.

Tangent at the origin: To know the nature of a multiple point it is necessary to find the tangent at that point.

The equation of the tangent at the origin can be obtained by equating to zero the lowest degree term in the equation of the curve.

3. **Point of intersection with co-ordinate axes:** By putting $y = 0$ in the equation of the curve we get the co-ordinate point of intersection with the x -axis.

Example. $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ put $y = 0$ we get $x = \pm a$. Thus $(a,0)$ and $(-a,0)$ are the co-ordinates of point of intersection.

By putting $x = 0$ in the equation of the curve, the co-ordinate of the point of intersection with the y -axis is obtained by solving the new equation.

4. **Region in which curve does not lie:** If the value of y is imaginary for certain value of x then the curve does not exist for such values.

Example: $y = \sqrt{x}(x + a)$. For $x < 0$, y is not real. So, no part of curve will lie in the region where $x < 0$.

5. Find all the asymptotes of the given curve including the asymptotes which are parallel to both x and y axes.
6. Intervals where curve is increasing, decreasing, maximum, minimum or point of inflexion.
i.e. find the intervals where $\frac{dy}{dx} > 0$ (increasing), $\frac{dy}{dx} < 0$ (decreasing).
Put $\frac{dy}{dx} = 0$ for the points where tangent is parallel to the x -axis.

If the curve traced appears to possess a point of inflexion, that point can be located by putting $\frac{d^2y}{dx^2}$ or $\frac{d^2x}{dy^2}$ equal to zero and solving the equation thus obtained.

7. **Table:** Prepare a table for certain values of x and y and draw the curve passing through them.

Example. Trace the curve $ay^2 = x^2(a - x)$.

Solution. We have,

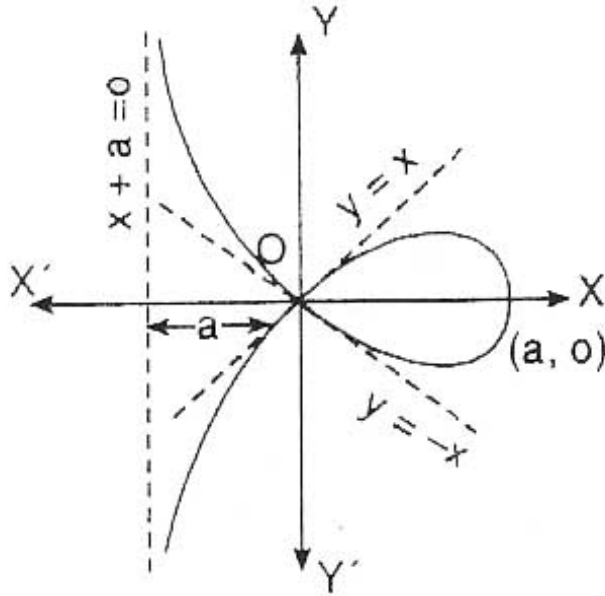
$$ay^2 = x^2(a - x) \quad (1)$$

1. **Symmetry:** Since the equation (1) contains only even powers of $y \Rightarrow$ it is symmetric about the x -axis.
It is not symmetric about y -axis since it does not contain even power of x .
2. **Origin:** Since constant term is absent, the curve passes through origin.
3. **Intersection with x -axis:** Putting $y = 0$ in (1), we get $x = a$.
Therefore, the curve cuts the x -axis at $(a, 0)$.

4. Tangent: The equation of the tangent at origin is obtained by equating to zero the lowest degree term of the equation (1).

$$\Rightarrow ay^2 = ax^2 \Rightarrow y^2 = x^2 \Rightarrow y = \pm x$$

Thus, there are two tangents $y = \pm x$ at the origin to the given curve.



Example. Trace the curve $y^2(a^2 + x^2) = x^2(a^2 - x^2)$.

Solution. Here, we have

$$y^2(a^2 + x^2) = x^2(a^2 - x^2).$$

1. Origin: The equation of the given curve does not contain constant term, therefore the curve passes through origin.
2. Symmetry about axes: The equation contains even powers of x as well as y , so the curve is symmetric about both the axes.
3. Point of intersection with x -axis: Putting $y = 0$ in the equation, we get

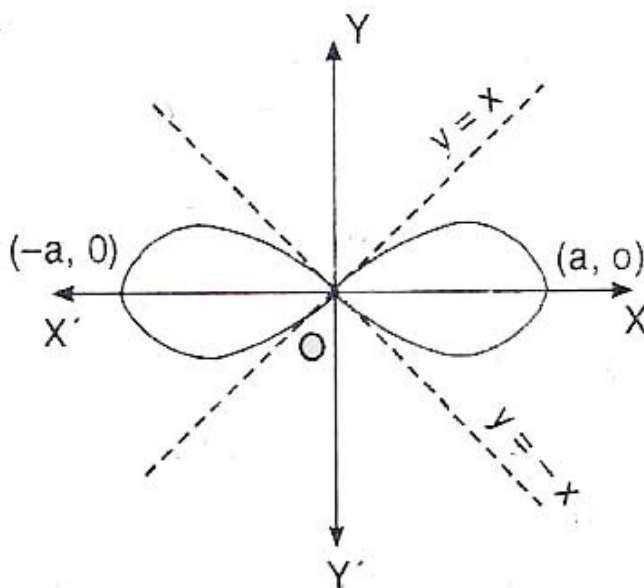
$$x^2(a^2 - x^2) = 0 \Rightarrow x = \pm a, 0.$$

4. Tangent at the origin: Equation of the tangent is obtained by equating to zero the lowest degree term

$$a^2y^2 - a^2x^2 = 0 \Rightarrow y = \pm x.$$

There are two tangents $y = x$ and $y = -x$ at the origin.

5. Node: Origin is the node, since there are two real and different tangents at the origin.
6. Region of absence of the curve: For values of $x > a$ and $x < -a$, y^2 becomes negative, hence the entire curve remains between $x = -a$ and $x = a$.



Reference Book: Differential Calculus, Dr. Gorakh Prasad. Pothishala Private LTD.